

The Credit Assignment Problem

Associative learning comes with another problem: When an animal gets food, there is never a single predictive cue beforehand but rather a whole swath of cues. If you pair a tap to the side of a slug with a shock, how does a slug's brain know to associate only the tap with the shock and not the many other sensory stimuli that were present, such as the surrounding temperature, the texture of the ground, or the diverse chemicals floating around the seawater? In machine learning, this is called the *credit assignment problem*: When something happens, what previous cue do you give credit for predicting it? The ancient bilaterian brain, which was capable of only the simplest forms of learning, employed four tricks to solve the credit assignment problem. These tricks were both crude and clever, and they became foundational mechanisms for how neurons make associations in all their bilaterian descendants.

The first trick used what are called *eligibility traces*. A slug will associate a tap with a subsequent shock only if the tap occurs one second before the shock. If the tap occurs two seconds or more before the shock, no association will be made. A stimulus like a tap creates a short eligibility trace that lasts for about a second. Only within this short time window can associations be made. This is clever, as it invokes a reasonable rule of thumb: stimuli that are useful for predicting things should occur *right before* the thing you are trying to predict.

The second trick was *overshadowing*. When animals have multiple predictive cues to use, their brains tend to pick the cues that are the strongest—strong cues *overshadow* weak cues. If a bright light and a weak odor are both present before an event, the bright light, not the weak odor, will be used as the predictive cue.

The third trick was *latent inhibition*—stimuli that animals regularly experienced in the past are inhibited from making future associations. In other words, frequent stimuli are flagged as irrelevant background noise. Latent inhibition is a clever way to ask, “What was different this time?” If a slug has experienced the current texture of the ground and the current temperature a thousand times but has never experienced a tap before, then the tap is far more likely to be used as a predictive cue.

The fourth and final trick for navigating the credit assignment problem was *blocking*. Once an animal has established an association between a

predictive cue and a response, all further cues that overlap with the predictive cue are blocked from association with that response. If a slug has learned that a tap leads to shock, then a new texture, temperature, or chemical will be blocked from being associated with the shock. Blocking is a way to stick to one predictive cue and avoid redundant associations.

The Original Four Tricks for Tackling the Credit Assignment Problem

ELIGIBILITY TRACES	OVERSHADOWING	LATENT INHIBITION	BLOCKING
Pick the predictive cue that occurred between 0 to 1 second <i>before</i> the event.	Pick the predictive cue that was the strongest.	Pick the predictive cue that you haven't seen before.	Stick to predictive cues once you have them and ignore others.

Eligibility traces, overshadowing, latent inhibition, and blocking are ubiquitous across Bilateria. Pavlov identified these in the conditional reflexes of his salivating dogs; they are found in the involuntary reflexes of humans; and they are seen in the associative learning of flatworms, nematodes, slugs, fish, lizards, birds, rats, and most every bilaterian in the animal kingdom. These tricks for navigating the credit assignment problem evolved as far back as the very first brains to make associative learning work.

These tricks are hardly perfect. In some circumstances, the best predictive cue may occur one minute before the event, not one second before. In other circumstances, the best predictive cue may be the weak cue, not the strong one. Over time, brains evolved more sophisticated strategies for solving the credit assignment problem (stay tuned for [breakthrough #2](#) and [breakthrough #3](#)). But the remnants of the first solutions—eligibility traces, overshadowing, latent inhibition, and blocking—still exist in modern brains. They are seen in our involuntary reflexes and in our most ancient brain circuits. Indeed, rats with their entire brains removed, with nothing left but their neural circuits in their spinal cord, still show latent inhibition, blocking, and overshadowing. Along with acquisition, extinction, spontaneous recovery, and reacquisition, this portfolio of tricks make up the foundation of the neural mechanisms of associative learning, mechanisms

that are embedded deep into the inner workings of neurons, neural circuits, and brains themselves.

The Ancient Mechanisms of Learning

For thousands of years, two groups of philosophers have been debating the relationship between the brain and the mind. One group, the dualists, like Plato, Aquinas, and Descartes, argue that the mind exists separately from the brain. The entities might interact with each other, but they are distinct; the mind is something beyond the physical. The materialists, like Kanada, Democritus, Epicurus, and Hobbes, argued that whatever the mind is, it is located entirely in the physical structure of the brain. There is nothing beyond the physical. This debate still rages in philosophy departments around the world. If you have made it this far in the book, I will assume you lean on the side of materialism, that you—like me—tend to reject nonphysical explanations for things, even the mind. But by siding with the materialists, we introduce several issues that, at first, are hard to explain physically, the most obvious being learning.

You can read a sentence once and then immediately repeat it out loud. If we stick to a materialist view, this means that reading this sentence instantaneously *changed something physical in your brain*. Anything that leads to learning causes physical reorganization of *something* in the 86 billion neurons in each of our heads. Keeping track of a conversation, watching a movie, and learning to tie your shoes all must change the physicality of our brains.

People have been speculating on the physical mechanisms of learning for thousands of years, and even the dualists pontificated on materialist explanations for learning. Plato believed that the brain was like a wax tablet in which perceptions left lasting impressions; he believed that memories were these impressions. Descartes argued that memories were formed through creating new “folds” in the brain, “not unlike the folds which remain in this paper after it has once been folded.” Others speculated that memories were persistent “vibrations.” These ideas were all wrong, although through no fault of their originators; at the time, no one understood even the basic building blocks of the nervous system, so they could not even begin to conceive of how learning might work.